

Notes: Farrokhi & Lashkaripour (Econometrica, 2025)

Can Trade Policy Mitigate Climate Change?

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1 Big picture

- **Question.** Can trade policy solve the free-riding problem in international climate policy?
- **Two policy proposals.** The paper evaluates two canonical ideas: carbon border taxes as a unilateral second-best policy, and a Nordhaus-style climate club that uses contingent trade penalties to induce compliance.
- **Main framework.** The authors build a multi-country, multi-industry general equilibrium trade model with global supply chains of carbon, fossil-energy production, final-good production, international trade, carbon taxes, and border taxes/subsidies.
- **Main theoretical contribution.** They derive formulas for unilaterally optimal carbon taxes, import tariffs, export subsidies, and climate-club penalties. The formulas internalize both terms-of-trade incentives and climate externalities.
- **Key result for unilateral border taxes.** Adding optimally designed carbon border taxes to unilaterally optimal domestic carbon taxes is mostly ineffective: it lowers global emissions by only 6.6%, while carbon taxes alone lower emissions by 5.4%. The incremental reduction from border taxes is only 1.3 percentage points.
- **Benchmark.** Globally cooperative carbon pricing would reduce global emissions by 41.0%, with a modest real-consumption loss and a positive climate-adjusted welfare gain.
- **Why border taxes are weak.** Carbon border taxes mainly hit traded goods, but much of global carbon is embedded in nontraded production and household energy use. Border taxes also trigger leakage through general equilibrium price responses.
- **Climate club result.** Climate clubs are far more powerful when trade penalties are used as off-equilibrium threats rather than as actual equilibrium taxes. Depending on the core coalition, a club can reduce global emissions by 13.4%, 18.6%, or 28.0%.
- **Best climate club.** If the EU, US, and China form the core, the maximal carbon-tax target that sustains universal membership is \$89/tCO₂, generating a 28.0% global emissions reduction, about 68% of the globally cooperative benchmark.
- **Economic takeaway.** Trade policy is weak as a substitute for carbon pricing, but strong as an enforcement mechanism for a climate agreement. The distinction is between using border taxes *on equilibrium* versus using trade penalties *off equilibrium* to deter free-riding.

2 Data and measurement

2.1 Countries, industries, and baseline year

- The quantitative economy is calibrated to 2014.
- The country set contains 19 regions: 13 large countries, the EU, Africa, and four residual regional aggregates.
- The industry set contains 18 industries: one composite energy industry plus 17 nonenergy industries.
- Trade, production, and expenditure data come from the Global Trade Analysis Project (GTAP) database.
- The baseline economy includes observed trade flows, production, expenditure, carbon emissions, tariffs, carbon taxes, and zero export subsidies.

Interpretation. The model is not a small theoretical example. It is a calibrated global economy with enough heterogeneity to capture cross-country asymmetries in output, emissions, trade exposure, and climate preferences.

2.2 Carbon accounting

- The paper counts carbon emissions where fossil energy is used by final-good producers and households.
- Production-related emissions make up roughly three-fourths of global emissions; households account for the remaining fourth.
- Most production-related emissions are embedded in goods that do not cross borders.
- Carbon intensity is measured as CO₂ emissions per dollar of output, normalized in the industry table and country table.

Interpretation. This accounting matters because border taxes operate through trade flows. If much carbon is in nontraded goods or domestic household energy use, border taxes cannot directly target a large part of emissions.

2.3 Key industry statistics

- Transportation is the largest source of industrial CO₂ emissions in Table I, at 27.3%.
- Metals account for 14.7%, other services for 14.5%, chemicals for 9.5%, and nonmetallic minerals for 8.6%.
- Highly traded industries are not necessarily the high-emission industries. Electronics and machinery has a 30.0% trade-to-GDP ratio but only 3.0% of industrial emissions.
- Carbon-intensive sectors include transportation, nonmetallic minerals, metals, chemicals, and other mining.

Interpretation. The mismatch between trade exposure and emissions is one reason carbon border taxes are weak. The sectors that are easiest to tax at the border are not always the sectors where emissions are concentrated.

2.4 Key country statistics

- China accounts for 26.7% of world CO₂ emissions in the sample; the US accounts for 17.2%; the EU accounts for 12.3%.
- The EU, US, and China together account for a large share of output and emissions, which is why adding China to a climate club core has a large effect.
- Country-level perceived disutility from carbon emissions, $\tilde{\delta}_i$, is calibrated from environmentally related taxes per unit of emissions, adjusted for country size.
- The $\tilde{\delta}_i$ values sum to the global social cost of carbon, calibrated at \$156.2 per tonne of CO₂ in 2014 dollars.

Interpretation. The model deliberately distinguishes actual carbon damage from the damage perceived or internalized by governments. This is what creates free-riding in the policy problem.

2.5 Elasticities and structural parameters

- Industry-level trade elasticities are estimated using a strategy resembling Caliendo and Parro.
- The estimates range across industries and average about 5.9.
- The preferred long-run energy demand elasticity is about 0.65.
- The carbon-reserve cost share in the energy industry is calibrated from the value-added share of natural resources in primary energy production.

Interpretation. Elasticities determine how trade flows, energy use, and emissions respond to taxes. The paper focuses on long-run policy outcomes, so long-run elasticities are appropriate.

3 Model and policy formulas

3.1 Prices and tax instruments

Country i can use three types of taxes:

- import taxes $t_{ji,k}$ on imports from origin j ;
- export subsidies $x_{ij,k}$ on exports to destination j ;
- carbon taxes $\tau_{i,k}$ on the carbon content of energy use.

The consumer price of variety (j, i, k) satisfies

$$\tilde{P}_{ji,k} = \frac{1 + t_{ji,k}}{1 + x_{ji,k}} P_{ji,k}. \quad (1)$$

The price of energy used in industry k is

$$\tilde{P}_{i,0,k} = \tilde{P}_{i,0} + \tau_{i,k}. \quad (2)$$

Interpretation. Import taxes and export subsidies create trade-price wedges. Carbon taxes create wedges in the user cost of energy. Together these instruments are sufficient to decentralize the relevant cooperative and noncooperative outcomes.

3.2 Consumption and trade shares

Consumers have indirect utility

$$V_i(E_i, \tilde{\mathbf{P}}_i), \quad (3)$$

where E_i is national expenditure and $\tilde{\mathbf{P}}_i$ is the vector of consumer prices. Expenditure shares are

$$\lambda_{ji,k} = \frac{\tilde{P}_{ji,k} C_{ji,k}}{\sum_n \tilde{P}_{ni,k} C_{ni,k}}, \quad \beta_{i,k} = \frac{\sum_n \tilde{P}_{ni,k} C_{ni,k}}{E_i}. \quad (4)$$

Interpretation. The policy formulas depend on expenditure shares because border taxes affect demand reallocations across origin countries and sectors.

3.3 Production, energy use, and emissions

Final-good output in country i , industry k , is produced with labor and energy:

$$Q_{i,k} = \bar{\varphi}_{i,k} F_{i,k}(L_{i,k}, Z_{i,k}). \quad (5)$$

Emissions in country-industry (i, k) are

$$Z_{i,k} = z_{i,k} Q_{i,k}, \quad Z_i = \sum_{k=1}^K Z_{i,k}, \quad Z^{global} = \sum_i Z_i. \quad (6)$$

Interpretation. Carbon policy changes both scale and technique: output can move across sectors and countries, while energy intensity can change within a sector.

3.4 Welfare with climate damage

Climate-adjusted welfare in country i is

$$W_i = V_i(E_i, \tilde{\mathbf{P}}_i) - \delta_i Z^{global}. \quad (7)$$

Interpretation. Each country suffers from global emissions, but a unilateral government internalizes only the damage to its own residents. This is the source of the carbon-pricing gap.

3.5 Unilaterally optimal policy

The unilaterally optimal domestic carbon tax is

$$\tau_i^* = \tilde{\delta}_i. \quad (8)$$

For nonenergy final-good imports, the optimal import tax has a terms-of-trade component and a carbon-border component:

$$1 + t_{ni,k}^* = (1 + \bar{t}_i) + \tau_i^* v_{n,k}. \quad (9)$$

Interpretation. The border-tax component taxes the carbon content per dollar of imported value. It is a second-best tool because the home country cannot directly tax foreign energy use.

3.6 Globally optimal policy

A global planner chooses taxes and transfers to maximize an international social welfare function. The globally optimal policy is

$$\tau_{i,k}^* = \sum_n \tilde{\delta}_n \equiv \tau^*, \quad t_{ij,k}^* = x_{ij,k}^* = 0. \quad (10)$$

Interpretation. The globally optimal carbon tax equals the global marginal damage from carbon. Border taxes are zero because direct carbon pricing is a better instrument once cooperation is possible.

3.7 The free-riding problem

The free-riding gap is

$$\tau_i^* = \tilde{\delta}_i < \tau^* = \sum_n \tilde{\delta}_n. \quad (11)$$

Interpretation. If all other countries impose the globally optimal carbon tax, country i still prefers to lower its own carbon tax to $\tilde{\delta}_i$. This strategic incentive generates under-pricing of carbon globally.

3.8 Climate club mechanism

A climate club specifies a core set of members and a target carbon tax τ^{target} . Members:

- raise their domestic carbon taxes to the target;
- impose zero border taxes on other members;
- impose optimal trade penalties on nonmembers.

Nonmembers retaliate against members and otherwise keep status quo policies.

Interpretation. The climate club works if penalties are strong enough to make membership individually optimal. If everyone joins, penalties are not used in equilibrium and free trade is preserved.

4 Identification and quantitative design

4.1 What is being identified or disciplined

- The model is disciplined by observed trade flows, expenditure shares, industry production, emissions, carbon intensity, tariffs, carbon taxes, energy cost shares, and estimated elasticities.
- The counterfactuals are not reduced-form treatment effects. They are quantitative general equilibrium simulations based on theoretically derived optimal policy formulas.
- The main objects are long-run changes in emissions, real consumption, and climate-adjusted welfare under alternative policy regimes.

Interpretation. The credibility of the exercise depends on whether the calibrated model captures the major channels through which border taxes and carbon taxes affect global production and emissions.

4.2 Why optimal policy formulas matter

- Brute-force optimal policy in a rich GE model is computationally difficult because all policy instruments affect all equilibrium variables.
- The paper uses a dual decomposition method that breaks the optimal policy problem into independent subproblems.
- This yields formulas for optimal taxes without directly computing complicated general equilibrium derivatives.

Interpretation. The formulas allow the authors to evaluate the *best possible* version of carbon border taxes and climate-club penalties, rather than an arbitrary or simplified version.

4.3 What the paper can and cannot prove

- The paper can quantify model-implied upper bounds for the effectiveness of two trade-policy solutions.
- The paper cannot prove that governments will adopt the optimal formulas or sustain a climate club politically.
- It abstracts from some enforcement, measurement, and political economy frictions in real-world carbon border systems.
- Results are long-run counterfactuals, not short-run transition estimates.

Interpretation. The central lesson is about the policy frontier: if even optimal border taxes barely reduce emissions, less-targeted real-world border taxes are unlikely to solve the climate problem by themselves.

5 Tables, formulas, and figures

The paper has no main empirical figures. The visual material below directly crops the original tables and key formula panels from the uploaded paper. Adjacent tables are grouped to save space and improve comparison.

5.1 Table I + Table II: Industry and country statistics

Read:

- Table I shows that emissions and trade exposure are poorly aligned. Transportation accounts for 27.3% of industrial emissions, but has only a 10.5% trade-to-GDP ratio.

- Metals and other services each account for roughly 14-15% of industrial emissions.
- Electronics and machinery is highly traded, with a 30.0% trade-to-GDP ratio, but has only 3.0% of emissions.
- Table II shows that China, the US, and the EU are the largest emissions and output players.
- China emits 26.7% of world CO₂ in the sample; the US emits 17.2%; the EU emits 12.3%.
- The calibrated disutility values are largest for the EU, Africa, China, and India.

Economic message. Border taxes have limited reach because the most emission-intensive activity is not always highly traded. The country table explains why the identity of climate-club core members matters.

TABLE I
INDUSTRY-LEVEL STATISTICS.

Industry	CO ₂ Emission (% of Total)	Trade-to-GDP Ratio	Carbon Intensity (v)	Energy Cost Share (α)	Trade Elasticity ($\sigma - 1$)
Agriculture	4.2%	8.9%	100.0	0.030	3.80
Other Mining	1.9%	28.9%	183.0	0.055	10.16
Food	3.3%	8.0%	45.9	0.015	3.80
Textile	1.9%	22.8%	59.7	0.021	4.25
Wood	0.5%	8.4%	61.0	0.026	6.50
Paper	2.1%	8.9%	125.9	0.061	6.55
Chemicals	9.5%	21.9%	179.5	0.062	8.60
Plastics	1.8%	13.5%	89.1	0.056	8.60
Nonmetallic Minerals	8.6%	6.0%	458.4	0.121	5.27
Metals	14.7%	14.6%	205.2	0.066	5.99
Electronics and Machinery	3.0%	30.0%	42.0	0.022	3.98
Motor Vehicles	1.2%	23.3%	34.0	0.014	4.88
Other Manufacturing	0.6%	21.5%	42.0	0.032	4.80
Construction	1.5%	0.6%	59.2	0.025	5.94
Wholesale and Retail	3.6%	2.4%	34.7	0.017	5.94
Transportation	27.3%	10.5%	498.3	0.171	5.94
Other Services	14.5%	3.1%	26.7	0.012	5.94

Note: This table shows for every of the 17 nonenergy industries the share from world industrial CO₂ emission (not including households' emission), world-level trade-to-GDP ratio, global average carbon intensity (CO₂ emissions per dollar of output) normalized by that of agriculture, energy cost shares reported as unweighted mean across countries within each industry, and estimated trade elasticities.

Table I: industry-level statistics

TABLE II
COUNTRY-LEVEL STATISTICS.

Country	Share from World			Carbon	
	Output	CO ₂ Emission	Population	Intensity (v)	Disutility (δ)
Australia	1.8%	1.2%	0.3%	147.5	1.5
EU	25.9%	12.3%	7.5%	100.0	53.2
Brazil	2.8%	1.6%	2.8%	135.3	6.0
Canada	1.9%	1.6%	0.5%	176.1	1.2
China	17.8%	26.7%	18.9%	378.1	20.9
Indonesia	1.0%	1.5%	3.5%	302.0	0.5
India	2.4%	6.4%	17.9%	620.4	12.5
Japan	6.2%	3.6%	1.8%	127.6	6.0
Korea	2.2%	1.7%	0.7%	188.7	3.2
Mexico	1.4%	1.4%	1.7%	218.8	0.3
Russia	1.9%	4.4%	2.0%	436.5	0.2
Saudi Arabia	0.4%	1.5%	0.4%	752.4	0.0
Turkey	1.0%	1.1%	1.1%	245.5	4.9
USA	20.4%	17.2%	4.4%	162.0	6.8
Africa	2.6%	3.6%	15.9%	285.3	22.2
RO Americas	3.0%	2.7%	4.1%	194.7	9.8
RO Asia and Oceania	5.1%	5.5%	11.8%	253.0	6.6
RO Eurasia	0.7%	2.2%	1.9%	671.6	0.1
RO Middle East	1.6%	3.9%	2.8%	494.9	0.3

Note: This table shows for every of the 19 regions (13 countries + the EU + Africa + 4 "Rest Of" regions as collection of neighboring countries), their share from world output, CO₂ emissions, and population, and carbon intensity (CO₂ emissions per dollar of output) normalized by that of the EU, and CPB-adjusted disutility from one tonne of CO₂ emission, which sum to the capital

Table II: country-level statistics

5.2 Formula panels: unilateral optimal policy versus global optimal policy

Read:

- The unilateral carbon tax equals domestic perceived marginal damage, $\tau_i^* = \tilde{\delta}_i$.
- Final-good import tariffs include a conventional trade-policy term and a carbon-border adjustment based on foreign carbon intensity $v_{n,k}$.
- Energy import and export policies additionally account for energy supply-chain linkages and reimported energy through the carbon supply chain.
- The global planner instead sets carbon taxes equal to the global marginal damage and sets border taxes/subsidies to zero.
- The free-riding gap is the difference between $\tilde{\delta}_i$ and $\sum_n \tilde{\delta}_n$.

Economic message. Unilateral border taxes are second-best substitutes for missing foreign carbon taxes. Under global cooperation, they disappear because direct carbon pricing dominates indirect trade interventions.

arbitrary tax shifter.

PROPOSITION 2: Country i 's unilaterally optimal policy consists of (i) uniform carbon taxes ($\tau_{i,k}^* = \tau_i^*$),

$$\tau_i^* = \tilde{\delta}_i \equiv \delta_i \bar{P}_i,$$

(ii) import tariffs and export subsidies on final goods ($k \geq 1$) that are unique up to a uniform and arbitrary tax-shifter, $\bar{t}_i \geq 0$, augmented by a carbon border adjustment based on the CO_2 content per unit value of imported goods $v_{n,k}$ (Equation (22)),

$$1 + t_{ni,k}^* = (1 + \bar{t}_i) + \tau_i^* v_{n,k},$$

$$1 + x_{in,k}^* = \frac{1 + \varepsilon_{in,k}}{\varepsilon_{in,k}} \sum_{j \neq i} \left[(1 + t_{ji,k}^*) \frac{\lambda_{jn,k}}{1 - \lambda_{in,k}} \right],$$

(iii) import tariffs and export subsidies on energy,

$$1 + t_{ni,0}^* = (1 + \bar{t}_i)(1 + \omega_{ni,0}) + \tau_i^* \sum_{\ell \neq i} \sum_{j \neq i} \left[\tilde{\psi}_{jn}^{(\ell,0)} \rho_{j\ell,0} \frac{\xi_\ell}{\bar{P}_{\ell,0}} \right],$$

$$1 + x_{in,0}^* = \frac{1 + \varepsilon_{in,0}}{\varepsilon_{in,0}} \sum_{j \neq i} \left[(1 + t_{ji,0}^*) \frac{\lambda_{jn,0}}{1 - \lambda_{in,0}} \right] - \left(\lambda_{in,0} + \tau_i^* \frac{\xi_n}{\bar{P}_{n,0}} \right) \frac{(1 + \bar{t}_i)}{\varepsilon_{in,0}},$$

where $\lambda_{in,0} = \frac{\sum_k a_{n,k} Y_{n,0} p_{n,0,k}}{\sum_k a_{n,k} Y_{n,0}}$ is the fraction of energy exports reimported via the carbon supply chain; $\omega_{ni,0} = \sum_{j \neq i} \tilde{\psi}_{jn}^{(i,0)} \rho_{ji,0}$ is the inverse export supply elasticity of energy (for flows from n to i), where $\tilde{\psi}_{jn}^{(i,0)} \equiv \frac{\phi_j}{1 - \phi_j} \psi_{jn}^{(i,0)}$ represents backward linkages in the energy sector;²⁴ λ and ρ represent international expenditure and sales shares (Equations (6) and (22)); ξ is the demand elasticity of composite energy input (Equation (23)), and e denotes the Marshallian demand elasticities (Equation (5)).

Unilaterally optimal policy formulas

$$\max_i \sum_j \theta_j \ln W_i(\bar{\mathbb{I}}) \text{ subject to general equilibrium Equations (1)–(19),}$$

where $W_i \sim V_i(E_i + \Delta_i, \bar{P}_i) - \delta_i \times Z^{(\text{global})}$ is country i 's climate-adjusted welfare under policy, with $\sum_i \Delta_i = 0$, and θ_i is country i 's weight in the planner's problem. The inclusion of income transfers is essential, as it separates redistribution, addressed via transfers, from climate-related externalities, addressed via taxes.

Capitalizing on a variation of Proposition 2, we derive the globally optimal policy in Supplemental Appendix C. The optimal policy from a global perspective involves carbon taxes that correct the worldwide externality of carbon emissions, along with zero trade taxes:²¹

$$\tau_{i,k}^* = \sum_j \tilde{\delta}_j \sim \tau^*, \quad t_{ji,k}^* = x_{ji,k}^* = 0 \quad (\forall i, j, k). \quad (26)$$

The globally optimal border taxes are zero because border taxes are an inefficient policy for reducing carbon emissions compared to directly targeted carbon taxes. In the unilateral case, carbon border taxes were justified since country i 's government could not directly tax foreign carbon inputs. This missing policy limitation no longer applies in the globally optimal context.

The *free-riding problem* stems from the gap between the unilaterally optimal and globally optimal carbon tax rates. Specifically,

$$\tau_i^* = \tilde{\delta}_i < \tau^* = \sum_n \tilde{\delta}_n.$$

Globally optimal carbon pricing and free-riding gap

5.3 Table III: Impact of noncooperative and cooperative tax policies

Read:

- Noncooperative carbon plus border taxes reduce global emissions by 6.6%, reduce real consumption by 0.5%, and reduce climate-adjusted welfare by 0.2%.
- Noncooperative carbon taxes without border taxes reduce global emissions by 5.4% and raise climate-adjusted welfare by 0.2%.
- Therefore, optimal border taxes add only about 1.3 percentage points of extra global emissions reduction.
- Globally cooperative carbon pricing reduces global emissions by 41.0%, reduces real consumption by only 0.6%, and raises climate-adjusted welfare by 1.1%.
- The difference between 6.6% and 41.0% is the quantitative scale of the free-riding problem.

Economic message. Carbon border taxes are weak as an indirect tax on global emissions. Direct cooperative carbon pricing is vastly more effective.

TABLE III
THE IMPACT OF NONCOOPERATIVE AND COOPERATIVE TAX POLICIES.

Country	Non-Cooperative						Globally Cooperative		
	Carbon + Border Tax			Carbon Tax					
	ΔCO_2	ΔV	ΔW	ΔCO_2	ΔV	ΔW	ΔCO_2	ΔV	ΔW
Australia	0.3%	-0.6%	-0.5%	1.9%	-0.0%	0.1%	-39.6%	-1.2%	-0.4%
EU	-22.2%	-0.3%	-0.0%	-21.2%	-0.0%	0.2%	-38.5%	-0.4%	1.7%
Brazil	-1.5%	-0.1%	0.3%	-1.0%	0.0%	0.3%	-39.4%	0.3%	2.6%
Canada	8.1%	-1.6%	-1.5%	3.5%	-0.1%	0.0%	-42.6%	-1.2%	-0.6%
China	-9.8%	-0.1%	0.1%	-8.3%	0.0%	0.1%	-39.0%	-1.7%	-0.6%
Indonesia	1.2%	-0.2%	-0.1%	2.4%	-0.0%	0.1%	-42.9%	-3.1%	-2.7%
India	-6.9%	-0.3%	0.5%	-5.3%	0.0%	0.7%	-44.0%	4.5%	10.8%
Japan	-2.1%	-0.3%	-0.1%	-0.6%	0.0%	0.1%	-39.1%	-1.5%	-0.5%
Korea	0.5%	0.3%	0.6%	0.9%	0.0%	0.2%	-39.9%	1.6%	3.2%
Mexico	4.3%	-1.3%	-1.2%	3.0%	-0.0%	0.0%	-41.5%	-1.3%	-1.1%
Russia	7.1%	-1.4%	-1.3%	3.5%	-0.2%	-0.2%	-43.8%	-0.0%	0.1%
Saudi Arabia	11.5%	-3.9%	-3.9%	4.8%	-0.6%	-0.6%	-45.8%	-0.6%	-0.5%
Turkey	-4.6%	-0.5%	0.3%	-0.0%	0.1%	0.8%	-39.1%	1.9%	7.6%
USA	-4.0%	-0.3%	-0.3%	-1.9%	0.0%	0.0%	-43.0%	-1.7%	-1.3%
Africa	-12.8%	-1.2%	0.1%	-10.2%	-0.1%	1.1%	-41.7%	8.4%	20.6%
RO Americas	-5.3%	-0.7%	-0.2%	-3.4%	-0.0%	0.4%	-41.5%	2.0%	5.6%
RO Asia	-5.5%	-1.0%	-0.9%	-0.9%	0.0%	0.2%	-40.6%	-0.9%	0.4%
RO Eurasia	2.0%	-1.1%	-1.1%	3.6%	-0.1%	-0.1%	-44.2%	-2.2%	-2.1%
RO Middle East	5.6%	-2.5%	-2.5%	3.9%	-0.3%	-0.3%	-43.3%	0.0%	0.2%
Global	-6.6%	-0.5%	-0.2%	-5.4%	-0.0%	0.2%	-41.0%	-0.6%	1.1%

Note: This table shows the change to CO_2 emissions, real consumption, and welfare for every country when moving from the baseline to noncooperative and cooperative equilibrium. In the baseline, each country's tariffs and carbon taxes are set at their applied rates in 2014 and export subsidies are zero. Optimal policy formulas for the noncooperative and cooperative outcomes are detailed in

Table III: noncooperative and globally cooperative outcomes

5.4 Table IV + Table V: Climate club accession and outcomes

Read:

- Table IV shows the accession path when the EU and US are the core and the carbon target is \$53/t CO_2 .
- Countries join over six rounds, and the process converges to universal membership.
- Table V reports maximal targets and emissions reductions for alternative core coalitions.
- EU-only core: target \$36/t CO_2 and 13.4% global emissions reduction.
- EU + US core: target \$53/t CO_2 and 18.6% reduction.
- EU + US + China core: target \$89/t CO_2 and 28.0% reduction.

Economic message. Climate clubs work because penalties are credible enough to induce membership. The larger the core coalition, the higher the sustainable carbon price and the larger the emissions reduction.

CLIMATE CLUB GAME WITH THE EU AND US AS CORE AND CARBON TAX
TARGET OF 53 (\$/t CO_2).

Core	EU, USA
Round 1	Brazil, Canada, Korea, Turkiye, RO Eurasia
Round 2	Russia, RO Americas
Round 3	Africa
Round 4	Japan, Mexico, Saudi Arabia
Round 5	China, Indonesia, RO Asia, RO Middle East
Round 6	Australia, India

Note: This table shows the convergence of our solution method via successive rounds to a club of all nations, for the case in which the EU and US are core members and the carbon tax target is at its maximal value of 53 \$/t CO_2 . A country unilaterally evaluates to join at each round given the club configuration at its previous round.

Core	Max Carbon Target (\$/t CO_2)	Reduction in Global CO_2
EU	36	-13.4%
EU + USA	53	-18.6%
EU + USA + China	89	-28.0%

Note: This table shows the climate club outcomes of the maximal carbon price target and the corresponding reduction in global CO_2 emissions for each scenario of the core countries

TABLE V

Table IV: climate club accession path

Table V: climate club outcomes

6 Short summary

- The paper asks whether trade policy can mitigate climate change by solving free-riding.
- It compares two trade-policy proposals: unilateral carbon border taxes and climate clubs with contingent penalties.
- The framework is a multi-country, multi-industry GE trade model with fossil energy, carbon supply chains, and climate externalities.
- The unilaterally optimal domestic carbon tax equals the damage perceived by the home government, not the global marginal damage.
- The unilaterally optimal border tax contains a terms-of-trade component and a carbon-border component.
- The globally optimal policy is direct carbon pricing at the global marginal damage level and zero border taxes.
- The model is calibrated to 2014 data from GTAP, emissions data, tariffs, carbon taxes, trade elasticities, and energy-demand elasticities.
- Border taxes are quantitatively weak: adding them to noncooperative carbon taxes reduces global emissions by only an extra 1.3 percentage points.
- The globally cooperative benchmark reduces global emissions by 41.0%.
- Climate clubs are more promising because penalties are used as threats, not as equilibrium trade barriers.
- With EU + US + China as the core, the climate club reaches a sustainable \$89/tCO₂ target and a 28.0% emissions reduction.
- The main message is that trade policy cannot substitute for global carbon pricing, but it can help enforce it.

7 Conclusion

7.1 Core conclusion

Farrokhi and Lashkaripour conclude that carbon border taxes are not enough to solve the climate free-riding problem. Even optimally designed border taxes deliver only a small fraction of the emissions reduction achieved by globally optimal carbon pricing. The climate club approach performs much better because it uses trade penalties to induce countries to adopt direct carbon pricing.

7.2 Economic intuition

Border taxes are indirect. They affect only traded goods, and even for traded goods they operate through prices, substitution, and general equilibrium leakage. Direct carbon taxes hit the actual externality: energy use and emissions. A climate club can preserve the superiority of direct carbon pricing by using trade penalties as a deterrent against free-riding.

7.3 Takeaway

The paper's policy lesson is sharp: trade policy is not a good substitute for climate policy, but it can be a useful enforcement device. The best use of trade penalties is to keep countries inside a cooperative carbon-pricing agreement, not to tax carbon indirectly through borders forever.